

Problem Statement #33 — Retail, E-Commerce & Finance

Micro-Lending & Peer Credit Risk Assessor

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Target Stakeholders / Actors	Borrower, Peer Lender		

A peer-to-peer micro-finance platform that calculates borrower risk scores using heuristic rules, generates structured repayment schedules, and triggers automated default warnings so that peer lenders can make informed lending decisions.

1. Requirements Table

1.1 Functional Requirements (FR-001 – FR-005)

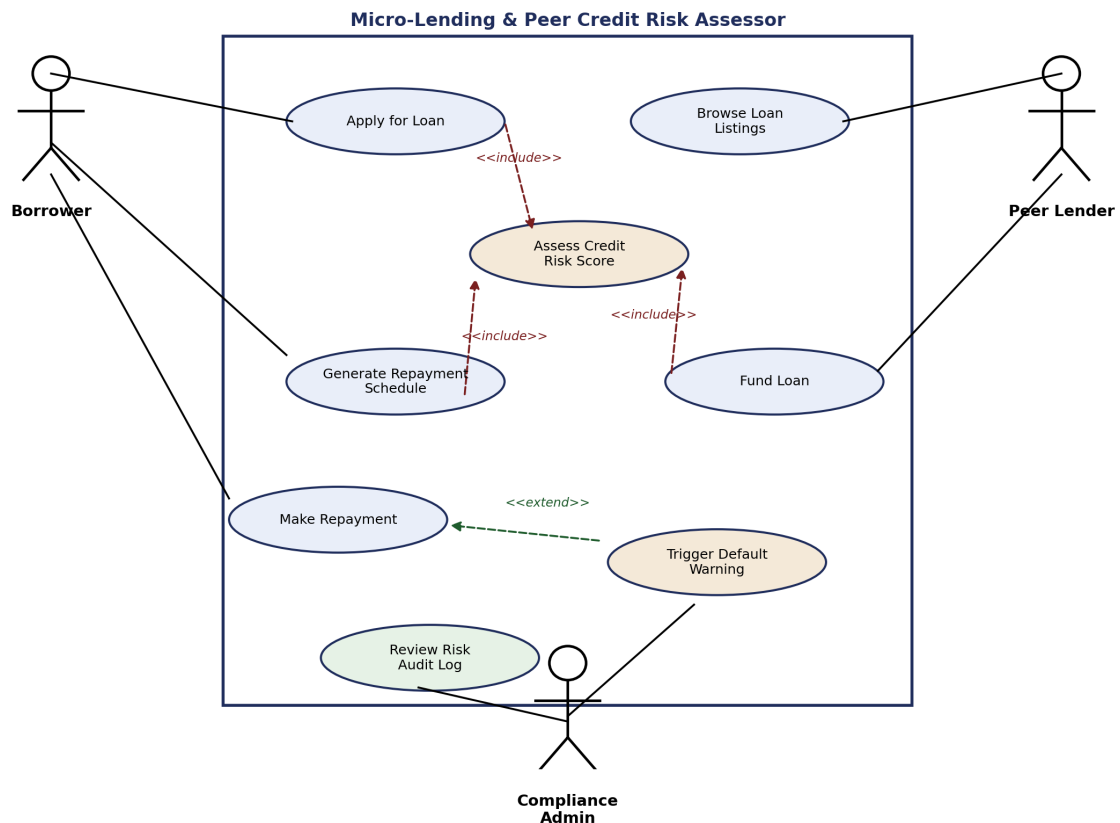
ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall compute a borrower credit risk rating (Low / Medium / High) based on debt-to-income ratio, repayment history, and employment stability indicators.	High	Pass: Risk tier accurately assigned for a complete profile. Fail: A loan is approved from an incomplete financial profile.	Risk scoring is the platform's core value proposition and gates every downstream lending decision.
FR-002	Functional	The system shall generate a structured EMI (equated monthly instalment) repayment schedule for every funded loan, including due dates and outstanding balance per instalment.	High	Pass: Schedule totals reconcile exactly with principal + interest. Fail: Any instalment date or amount is missing or inconsistent.	Borrowers and lenders both need a single source of truth for repayment obligations.
FR-003	Functional	The system shall allow a peer lender to browse open loan requests filtered by risk tier, loan amount, and tenure before choosing to fund one.	Medium	Pass: Listing results match the filters applied within 2 seconds. Fail: A filtered result includes a loan outside the selected risk tier.	Lenders need transparent, filterable visibility into risk before committing capital.
FR-004	Functional	The system shall process a peer lender's loan-funding transaction and update the loan status to 'Funded' only once the full requested principal has been committed.	High	Pass: Loan status flips to Funded exactly at 100% commitment. Fail: Status changes with partial funding, or funds are committed to a withdrawn listing.	Prevents over-funding or under-funding disputes between multiple peer lenders.
FR-005	Functional	The system shall automatically trigger a default warning notification to the peer lender and compliance admin when a borrower misses a scheduled repayment by more than 7 days.	High	Pass: Warning is dispatched within 1 hour of the 7-day threshold being crossed. Fail: No warning is generated for a missed payment past the threshold.	Early default warnings let lenders and compliance staff intervene before losses compound.

1.2 Non-Functional Requirements (NFR-001 & NFR-002)

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	Loan repayment transactions and EMI distribution must execute inside database ACID transactions to prevent double-spending.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Financial correctness cannot be sacrificed for speed; concurrent repayments must never corrupt balances.
NFR-002	Confidentiality & Data Protection	Borrower financial profile data (income, repayment history, risk score) must be encrypted at rest and in transit, and accessible only to the borrower, the funding lender, and compliance admin.	High	Pass: Penetration test confirms data is unreadable outside authorized roles; Fail: Any unauthorized role can read financial fields in plaintext.	Micro-lending profiles are sensitive financial data; a breach directly harms borrower trust and regulatory standing.

2. UML Use-Case Diagram

Primary actors: **Borrower** and **Peer Lender**. Secondary actor: **Compliance Admin** (oversees default warnings and risk audit logs). The diagram below models all actors, the primary use cases, one «include» relationship chain (Apply for Loan, Generate Repayment Schedule, and Fund Loan all include Assess Credit Risk Score, since risk scoring is mandatory sub-behaviour for each), and one «extend» relationship (Trigger Default Warning optionally extends Make Repayment, firing only on the missed-payment condition).



Actors & Use Cases Summary

Actor	Type	Use Cases Involved
Borrower	Primary	Apply for Loan, Generate Repayment Schedule, Make Repayment
Peer Lender	Primary	Browse Loan Listings, Fund Loan
Compliance Admin	Secondary	Trigger Default Warning (notified), Review Risk Audit Log

3. Use-Case Flow Specification

Core Use Case: Apply for Loan (includes Assess Credit Risk Score)

Actor(s):

Borrower (primary), System Risk Engine (via «include»)

Preconditions:

1. The borrower has a registered, verified platform account.
2. The borrower has submitted employment and income details required for the risk profile.
3. The borrower has no existing loan application currently in 'Pending Review' status.

Postconditions:

1. A new loan request record exists with a computed risk tier (Low / Medium / High) and a status of 'Open for Funding' or 'Rejected'.
2. If approved, the request becomes visible to peer lenders in the loan listings.
3. The borrower receives a confirmation notification with the assigned risk tier.

Main Success Scenario:

1. The borrower selects “Apply for Loan” and enters the requested loan amount, tenure, and purpose.
2. The system validates that all mandatory profile fields (income, employment status, repayment history) are complete.
3. The system «includes» the Assess Credit Risk Score use case, computing debt-to-income ratio, repayment history, and employment stability indicators.
4. The system assigns a risk tier (Low / Medium / High) to the application.
5. The system generates a preliminary structured repayment schedule based on the requested amount, tenure, and the assigned risk tier's interest rate.
6. The system sets the loan status to “Open for Funding” and publishes it to the peer lender listings.
7. The system displays a confirmation to the borrower showing the risk tier, proposed EMI schedule, and expected funding timeline.

Alternate Flow — A1: Incomplete Financial Profile

1. At step 2, the system detects that one or more mandatory profile fields are missing or incomplete.
2. The system halts the application and displays which fields are missing (e.g. income proof, employment status).
3. The borrower is redirected to the profile completion screen instead of the risk assessment step.
4. Once the borrower resubmits a complete profile, the use case resumes from step 2 of the Main Success Scenario.

Deliverables included in this document: (1) Complete Requirements Table – 5 Functional Requirements and 2 Non-Functional Requirements, each with ID, Type, Description, Priority, Acceptance Criteria, and Rationale; (2) UML Use-Case Diagram with all actors, primary use cases, one «include» and one «extend» relationship; (3) Use-Case Flow Specification for the core “Apply for Loan” use case with Preconditions, Postconditions, Main Success Scenario, and one Alternate Flow.